

SDR Spectrum Analyzer User Guide

Product: SDR Spectrum Analyzer **Publisher:** Mattino 507 Software **Version:** 1.0 **Last updated:** September 2026

1. Introduction

SDR Spectrum Analyzer is a Windows desktop application for viewing and analysing RF spectrum data with tested compatible USB software-defined radio (SDR) hardware. It presents frequency-domain traces, markers, channel analysis, configurable acquisition settings, segmented sweeps, and CSV export.

It is a spectrum visualisation and analysis application, not a general-purpose radio receiver or audio-demodulation application.

2. Measurement scope and important limitations

V1.0 normally displays relative signal levels in **dBFS**: decibels relative to the digital full-scale level of the SDR. dBFS is not the same as RF input power in dBm.

Displayed values depend on the SDR tuner and ADC response, selected gain, frequency, sample rate, connected cables and attenuators, and the RF environment. A value nearer 0 dBFS is stronger, but values too near 0 dBFS can clip the ADC and distort the spectrum.

The application includes a channel-power calibration workflow for a matching measurement setup and reference instrument. Such a calibration is conditional on its saved setup; it is not a universal device calibration. Do not treat V1.0 as a calibrated laboratory spectrum analyser, a precision power meter, or a guarantee of absolute dBm accuracy.

3. System and hardware requirements

- A supported Windows desktop environment. The current MSIX candidate targets Windows Desktop with a minimum version of Windows 10 2004.
- Tested RTL2838-based USB SDR hardware (VID:PID `0BDA:2838`) with an R820T-series tuner.
- A functional Microsoft WinUSB binding on the device's Interface 0 (`MI_00`).

Only the tested RTL2838/R820T-series path is claimed. Compatibility with all RTL-SDR devices, RTL2832U devices, clones, or other tuner families is not claimed.

4. Installation

Install SDR Spectrum Analyzer using the release package supplied for the chosen release channel.

Application installation and SDR driver preparation are separate. The application does not install, download, update, replace, or otherwise modify drivers. Compatible hardware must already have the required Windows driver configuration before it can be accessed.

The production WinUSB signing and distribution architecture remains subject to Microsoft guidance. This guide deliberately does not provide a driver-download, certificate, Test Mode, Zadig, or recovery procedure. Detailed driver troubleshooting belongs to future supported troubleshooting material.

5. Getting started

1. Start SDR Spectrum Analyzer.
2. Connect tested compatible SDR hardware whose Interface 0 is already ready for WinUSB.
3. In **Device & Acquisition**, select **Refresh** and select the listed device.
4. Select **Connect**. The diagnostics area reports the detected device and readiness result.
5. Set frequency, span, acquisition mode, sample rate, FFT points, window, gain, frequency correction, and sweep averaging as needed.
6. Select **Single Sweep** for one completed acquisition or **Continuous Run** for repeated acquisition.
7. Inspect the trace, markers, channel range, and status information.
8. Select **Stop** when you need to end a running acquisition.

6. Main window overview

[Screenshot: Main application window]

The main window contains these areas:

- **Device & Acquisition:** hardware selection and capture settings.
- **Display & Markers:** scaling, traces, markers, path offset, and channel-power controls.
- **Spectrum display:** traces, axes, channel boundaries, and marker readouts.
- **Output & Diagnostics:** timestamped user-facing status and diagnostic messages.
- **Status bar:** device/acquisition state and spectrum or channel context.
- **Menu and toolbar:** shortcuts to the commands described below.

The left-side settings panes can be docked and may scroll when their available height is limited. **Restore Default Layout** restores the application toolbar and dock-pane layout.

7. Device and acquisition

Device controls

- **Device list:** lists enumerated SDR devices.
- **Refresh:** enumerates supported devices and reports readiness.
- **Connect:** opens the selected listed device.
- **Disconnect:** closes the current device connection.
- **Check Connection:** verifies an existing connection.

The readiness message distinguishes a missing device, an unavailable WinUSB Interface 0, a ready interface, and a backend access failure. A ready interface is still required before connection and capture can succeed.

Frequency and acquisition controls

- **Frequency unit:** Hz, kHz, MHz, or GHz for frequency entry.
- **Center / Span** or **Start / Stop:** the two frequency-entry modes. Use **Center+SPAN** to switch between them.
- **Channel BW MHz:** channel integration width. Applying it uses the current display centre as the channel centre.
- **Acq Mode:** fixed-span acquisition or segmented sweep.
- **Sample rate:** 1.024, 2.048, or 2.4 MS/s.
- **FFT Points:** 1024, 2048, 4096, 8192, or 16384.
- **Window:** Hann or Blackman-Harris.
- **Gain:** requested tuner gain from the available application list. The device selects the closest supported tuner step.
- **Freq Corr / PPM:** integer frequency correction from -200 to +200 PPM. Zero disables correction.
- **Sweep Avg:** 1, 4, 8, 16, or 32 completed sweeps.

Capture-affecting settings, data-source changes, and configuration load/save are unavailable while an acquisition is active. Stop first, then apply the change.

8. Acquisition modes

Fixed span

Fixed span is limited to the usable instantaneous bandwidth of the selected sample rate and FFT configuration. The application excludes edge and DC-affected bins from the usable stitched data area.

Segmented sweep

For a wider span, SDR Spectrum Analyzer retunes through sequential frequency segments and stitches their usable FFT bins into the displayed range. It does **not** acquire the entire displayed span simultaneously.

Each segment uses four FFT frames internally and averages their linear power. **Sweep Avg** then averages corresponding bins across complete sweeps. Higher averaging reduces variation but increases settling time and slows response to changing signals.

9. Running measurements

- **Continuous Run** starts repeated device acquisition or repeated Demo Data updates.
- **Single Sweep** completes one acquisition sweep and then stops.
- **Stop** requests that the current acquisition or demo run stop. The last published result remains displayed.
- **Clear Max Hold** clears the maximum-hold data collected so far.

When device data is active, acquisition requires a connected and usable SDR. The output area reports connection, capture, stop, and error status.

10. Spectrum display and traces

[Screenshot: Spectrum display and traces]

The plot shows frequency horizontally and dBFS by default vertically. If a matching saved channel calibration is active for the current device and setup, eligible channel and trace readouts can use dBm; otherwise they remain dBFS.

- **Current / Average Trace** shows the current processed result or the selected sweep average.
- **Max Hold** retains the highest observed value at each frequency bin until cleared.
- **Frozen Reference** shows a separately frozen trace for visual comparison. **Freeze Curr as Ref** captures the current trace and its original frequency range.
- **Ref Level dBFS** sets the top value of the display scale. It does not change hardware gain or acquired samples.
- **dB per** selects the vertical division size; **Apply Scale** applies the entered scale.
- **Auto Scale** sets a practical scale from the current trace.
- **Path Offset** is a non-negative display and calibration-path compensation value. Apply it after entering the known path loss or attenuation.

Stable production colours are green for Current / Average, yellow for Max Hold, and blue for Frozen Reference.

11. Markers and measurements

[Screenshot: Markers and channel analysis]

The plot has two markers, **M1** and **M2**.

- Left-click a valid plotted point to place M1.
- Shift + left-click a valid plotted point to place M2.
- Drag an M1 or M2 dashed line horizontally to move it.
- **M1→A** moves M1 to the highest valid displayed trace point.
- **M1→C** moves M1 to the display centre.
- **C→M1** changes the display centre to M1. If capture is running, the application stops and resumes it after the centre change.

Marker readouts show the marker frequencies and levels. Delta readout shows the frequency and level difference between M2 and M1.

Channel power and integration range

The channel range is centred on the current display centre and uses **Channel BW MHz**. The plot draws its range boundaries. A channel result becomes available only after a complete, valid sweep covering that range and the selected average count.

The **Ref Chan Power dBm** field and **Calibrate** button are for a reference-instrument calibration workflow. Calibration requires device data, a live connected device, a valid completed channel result, no clipping, a matching current capture setup, a valid path offset, and a reference input from -160 to +30 dBm. **Clear Calib** removes the saved channel calibration record.

12. Frequency and span controls

Use the selected frequency unit when entering values. **Center+SPAN** switches the two entry fields between:

- centre frequency plus span; and
- start frequency plus stop frequency.

The valid display range is 24 MHz to 1766 MHz, with a minimum span of 10 kHz. Changing range clears the active device snapshot; in Demo Data mode it produces a new generated view. Wider spans use segmented acquisition as described in [Acquisition modes](#).

Use the mouse wheel over the plot to zoom around the pointer. Drag an empty plot area to pan. These interactions are unavailable while a hardware capture is busy.

13. Data source

The **Data Source** submenu offers:

- **Device Data**: spectrum data acquired from a connected supported SDR.
- **Demo Data (not measured)**: generated demonstration data.

Demo Data does not access USB and cannot be used for RF measurement or calibration. It is intended for learning the user interface and display behaviour.

14. Configuration files

File > Save Configuration writes an `.cfg` text file selected by the user. **File > Load Configuration** reads a compatible `.cfg` file after acquisition has stopped.

The current format is versioned as `SDR_SA_CONFIG=1` and stores capture settings, display scale, trace visibility, channel width, path offset, data source, frequency-entry mode/unit, and an optional reference channel power. It is an application configuration file; do not assume it is a general interchange format or manually edit it unless you can validate the result in the application.

15. Exporting data

Export CSV is available when a spectrum trace exists. The application asks for a user-selected `.csv` location.

The CSV includes comment metadata for the data source, dBFS unit, capture settings where a device snapshot exists, channel result status, and path offset. Its table columns are:

```
frequency_hz,power_dbfs,valid
```

`valid` identifies bins containing usable plotted data. For device data, the file may also include a calibrated `channel_dbm` metadata field only when the current saved calibration matches the snapshot. The primary per-bin values remain dBFS.

16. Output and diagnostics

The **Output & Diagnostics** pane records timestamped messages for application readiness, device refresh, driver readiness, connection, acquisition start/stop, configuration actions, export, and user-visible errors.

Use it when reporting a problem. It is not a supported interface for internal driver installation, certificate setup, or engineering recovery procedures.

17. Status bar

The status bar presents a concise device and acquisition state plus spectrum context. Depending on the current data, it can include the data source, single/continuous or stopped state, channel integration range, acquisition mode, FFT bin spacing, ENBW, gain, path offset, calibration state, and sweep progress.

The displayed `uncalibrated` or 未标定 status means that the requested or active gain is not represented by a matching saved calibration; it does not by itself indicate a device failure.

18. Menus and toolbar reference

File

- **Load Configuration** and **Save Configuration**: see [Configuration files](#).
- **Exit**: closes the application.

Device

- **Refresh Devices**, **Connect Selected Device**, **Disconnect Device**, and **Check Connection**: see [Device and acquisition](#).

Run & Frequency

- **Continuous Run**, **Single Sweep**, **Stop**, **Clear Max Hold**, **Center+SPAN**, and **Export CSV**: see [Running measurements](#) and [Frequency and span controls](#).
- **Data Source**: see [Data source](#).
- **Restore Default Layout** restores the toolbar and dock panes.

View

- **Auto Scale**: see [Spectrum display and traces](#).

Help

- **User Guide** currently displays a short in-application usage summary.
- **Window Comparison and Selection** displays guidance for Hann and Blackman-Harris windows.
- **About** displays product identity, version, publisher, the third-party-notices reminder, and the local privacy statement.

The toolbar provides text buttons for Continuous Run, Single Sweep, Stop, Clear Hold, Export CSV, Center+SPAN, Apply Settings, and Restore Default Layout.

19. Window functions and FFT concepts

The FFT converts each captured time-domain I/Q block into frequency bins. More FFT points provide finer bin spacing at a given sample rate, while sample rate and the selected span determine the usable frequency coverage.

- **Hann** has approximately 1.50 FFT-bin ENBW and is suited to general viewing and ordinary channel-power work.
- **Blackman-Harris** has approximately 2.00 FFT-bin ENBW and much lower sidelobes, which can help when weak signals are near strong carriers.

The channel-power calculation applies ENBW correction. After changing a window, acquire again and recalibrate before relying on a calibrated channel result.

20. Measurement accuracy and interpretation

Use dBFS for relative spectrum interpretation unless the application explicitly shows a matching calibrated result. Device gain, PPM correction, and path offset have different purposes:

- **Gain** requests a tuner gain setting; the backend uses the closest supported step. It is not a power calibration.
- **Frequency correction (PPM)** compensates systematic oscillator frequency error. It does not change gain or power.
- **Path Offset** accounts for a known RF-path loss or attenuation in display and calibration context.
- **Channel power** integrates the configured channel range only after a valid completed sweep. It remains dBFS without matching calibration.
- **ENBW** affects broadband power interpretation and is accounted for by the channel calculation.

Use known references, matching settings, appropriate attenuation, and a stable measurement environment when performing any conditional calibration. Recheck calibration after changing device, gain, frequency-related conditions, window, or relevant measurement settings.

21. Troubleshooting

- **No device listed:** confirm that tested hardware is connected, then select Refresh.
- **WinUSB driver unavailable:** the device is present but Interface 0 is not ready for the Mattino backend. The application will not change a driver. Use only an approved production support path when one is published.
- **Unable to connect:** refresh the list, confirm readiness, and ensure another application is not holding the device.
- **Acquisition does not start:** stop any active acquisition, review the Output & Diagnostics message, and check valid frequency, span, channel bandwidth, FFT, sample-rate, gain, and PPM inputs.
- **Unexpected trace or channel result:** check clipping, selected data source, gain, path offset, window, averaging, marker/channel range, and whether a calibration actually matches the current device and setup.
- **USB device stops responding:** disconnect and reconnect through the application where possible, then refresh. Do not use engineering certificate or driver procedures as an end-user workaround.

22. Getting support

Technical support for SDR Spectrum Analyzer is available through the public product support site:

<https://wangyaoyang.github.io/mattino507-support/>

The support site provides support information and access to the public GitHub support-request system.

When requesting support, include the application version, Windows version, device readiness state, tested hardware identity, relevant diagnostic messages, and a screenshot when useful.

Do not post passwords, credentials, product keys, private certificates, personal addresses, or other sensitive personal information in a public request.

23. Privacy and license

Refer to the separately published Privacy Policy and End User License Agreement for the current release.

The current versions are available through the SDR Spectrum Analyzer product support site.

Do not duplicate Privacy Policy or EULA terms in this User Guide.

24. Third-party components

The commercial distribution may include third-party open-source/runtime components. Applicable notices and licence texts are supplied with the product in `THIRD_PARTY_NOTICES.txt` and its accompanying licence files. The commercial V1.0 package dynamically links libusb and includes Apache-2.0 attribution for Google Radio Receiver-derived material; it does not include or depend on the Legacy Osmocom rtl-sdr runtime.